

Name: **SOLUTIONS**

GCSE Maths Grade 1 Questions Calculator Pack D



This worksheet consists of the first two pages of GCSE maths past papers and represents roughly the equivalent work of grade 1 maths.

This table shows you which papers these questions came from.

Pages	The First Two Pages of
2 - 3	June 2017 Paper 3
4 - 5	Specimen 2 Paper 3
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Answer ALL questions.

Write your answers in the spaces provided.

You must write down all the stages in your working.

- 1 The table shows the lengths of five rivers.

River	Length (km)
Trent	297
Don	112
Severn	354
Thames	346
Mersey	113

- (a) Write down the rivers in order of length.

Start with the shortest river.

Don, Mersey, Trent, Thames, Severn
112 113 297 346 354

(1)

Ami says,

“The River Thames is more than three times as long as the River Don.”

- (b) Show that Ami is correct.

$$112 \times 3 = 336$$

$$346 > 336$$

Yes he is correct

(1)

(Total for Question 1 is 2 marks)

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- 2 Cups are sold in packs and in boxes.

There are 12 cups in each pack.

There are 18 cups in each box.

Alison buys p packs of cups and b boxes of cups.

Write down an expression, in terms of p and b , for the total number of cups Alison buys.

$$12 \times p + 18 \times b$$

$$12p + 18b$$

(Total for Question 2 is 2 marks)

- 3 Here are four digits.

5 6 1 9

- (i) Write down the smallest possible two digit number that can be made with two of the digits.

15

(1)

- (ii) Write down the three digit number closest to 200 that can be made with three of the digits.

196

(1)

(Total for Question 3 is 2 marks)



Answer ALL questions.

Write your answers in the spaces provided.

You must write down all the stages in your working.

- 1 Change 4500 g to kg.

$$4500\text{g} = 4.5\text{kg} \quad \dots\dots\dots 4.5 \text{ kg}$$

$\div 1000$ (Total for Question 1 is 1 mark)

- 2 Write 0.19 as a fraction.

$$0.19 = \frac{19}{100}$$

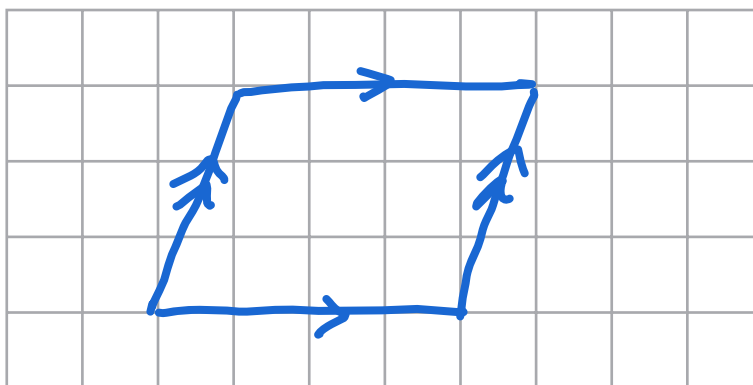
(Total for Question 2 is 1 mark)

- 3 Write down an even number that is a multiple of 7

$$7, \textcircled{14}, 21, \textcircled{28}, 35, \textcircled{42} \quad \dots\dots\dots \text{e.g. } 14$$

(Total for Question 3 is 1 mark)

- 4 On the grid, draw a parallelogram.



(Total for Question 4 is 1 mark)

- 5 Write
- $\frac{3}{5}$
- as a percentage.

$$\frac{3}{5} = \frac{6}{10} = \frac{60}{100} \quad \dots\dots\dots 60 \%$$

(Total for Question 5 is 1 mark)

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- 6 Coffee is sold in jars.
There are 200g of coffee in each jar.

Ben makes 8 cups of coffee each day.
He thinks he uses 2g of coffee to make each cup of coffee.

Ben wants to buy enough coffee for 28 days.

- (a) How many jars of coffee does Ben need to buy?

$$2g \times 8 = 16g \text{ per day}$$

$$28 \times 16 = 448g \text{ needed}$$

$$2 \text{ jars} = 400g$$

$$3 \text{ jars} = 600g$$

3

(3)

Ben finds that he uses 2.5g of coffee to make each cup of coffee.

- (b) How does this affect the number of jars of coffee he needs to buy?
You must give a reason for your answer.

$$2.5 \times 8 = 20g \text{ per day}$$

$$28 \times 20 = 560g \text{ needed.}$$

3 jars will still be enough.

(2)

(Total for Question 6 is 5 marks)

- 7 Write down three different factors of 18 that add together to give a prime number.

Factors of 18

1	18
2	9
3	6

$$2 + 6 + 9 = 17$$

or

$$3 + 6 + 2 = 11$$

or

$$2 + 3 + 18 = 23$$

or

$$2 + 9 + 18 = 29$$

(Total for Question 7 is 2 marks)

Answer ALL questions.

Write your answers in the spaces provided.

You must write down all the stages in your working.

- 1 Write the number 5689 correct to the nearest thousand.

6000

(Total for Question 1 is 1 mark)

- 2 Work out $\frac{30 + 12}{5 + 3}$

$$\frac{42}{8} =$$

5.25

(Total for Question 2 is 1 mark)

- 3 Work out the reciprocal of 0.125

$$0.125 = \frac{1}{8}$$

$$\left(\frac{1}{8}\right)^{-1} = 8$$

(Total for Question 3 is 1 mark)

- 4 Here is a list of numbers.

1

2

5

6

12

From the list, write down

- (i) a multiple of 4

4, 8, 12, 16

12

- (ii) a prime number

2, 3, 5, 7

2 or 5

(Total for Question 4 is 2 marks)



- 5 There are 1.5 litres of water in a bottle.

There are 250 millilitres of water in another bottle.

Work out the total amount of water in the two bottles.

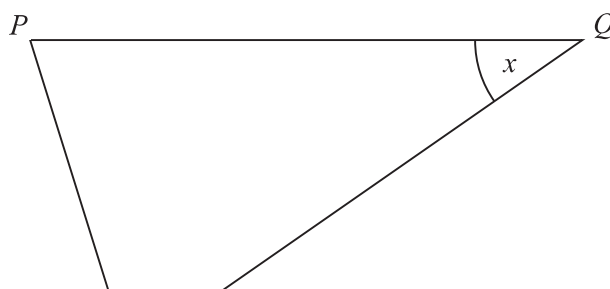
$$\begin{aligned} 1.5 \text{ litres} &= 1500 \text{ mL} \\ 0.25 \text{ litres} &= \underline{250 \text{ mL}} \\ 1.75 \text{ litres} &= 1750 \text{ mL} \end{aligned}$$

$$1.75 \text{ litres} \text{ or } 1750 \text{ mL}$$

(Total for Question 5 is 3 marks)

- 6 Here is a trapezium.

This diagram is accurately drawn.



- (a) Measure the length of the line PQ .

7cm
(1)

- (b) Measure the size of the angle marked x .

35°
(1)

(Total for Question 6 is 2 marks)



S 4 9 8 1 9 A 0 3 2 0

Answer ALL questions.

Write your answers in the spaces provided.

You must write down all the stages in your working.

- 1 Write 2148 correct to the nearest 100

2100

(Total for Question 1 is 1 mark)

- 2 (a) Simplify
- $8x - 3x + 2x$

$$8x - 3x = 5x$$

$$5x + 2x = 7x$$

(1)

- (b) Simplify
- $4y \times 2y$

$$8y^2$$

(1)

(Total for Question 2 is 2 marks)

- 3 There are 6760 people at a rugby match.
 3879 of the people are men.
 1241 of the people are women.
 $\frac{1}{4}$ of the children are girls.

$$3879 + 1241 = 5120 \text{ adults}$$

$$6760 - 5120 = 1640 \text{ children}$$

Work out how many boys are at the rugby match.

$$1640 \times \frac{1}{4} = 410 \text{ girls}$$

$$\text{Boys} = 1640 - 410 = 1230$$

1230

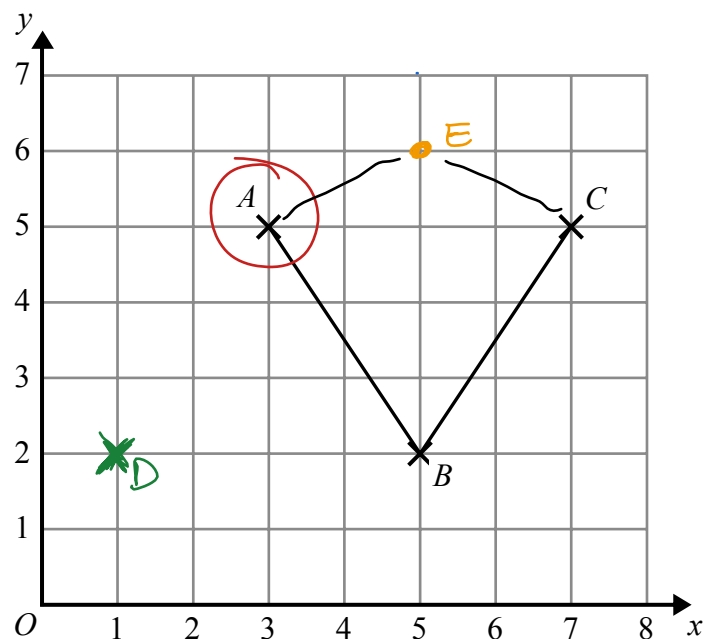
(Total for Question 3 is 3 marks)

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- 4 Here is a grid showing the points A , B and C .



- (a) Write down the coordinates of the point A .

$(3, 5)$

(1)

- (b) On the grid, mark with a cross ($\times$) the point $(1, 2)$.
Label this point D .

(1)

- (c) On the grid, mark with a cross ($\times$) a point E , so that the quadrilateral $ABCE$ is a kite.

(1)

(Total for Question 4 is 3 marks)

- 5 Faiza buys

one magazine costing £2.30
one paper costing 92p
two identical bars of chocolate

Faiza pays with a £5 note.
She gets 40p change.

Work out the cost of **one** bar of chocolate.

$$£5 - 40p = £4.60$$

$$2.30 + 0.92 = 3.22$$

$$4.60 - 3.22 = £1.38$$

$$£1.38 \div 2 = 69p$$

(Total for Question 5 is 3 marks)